Vector Addition — Resultant of Vectors

Problem Set with Detailed Solutions · Class 11 — JEE Foundation

Instructions: For each problem find the resultant — its magnitude, and its direction wherever the question asks for it. Use the parallelogram / triangle law of vector addition (no components or axes). Angles are measured between the vectors when drawn from a common point. Take $\sqrt{2} \approx 1.41$, $\sqrt{3} \approx 1.73$ where needed. (26 problems)

Problems

1. Two forces of 3 N and 4 N act at a point, at right angles (90°) to each other. Find the magnitude of their resultant.

2. Two vectors, each of magnitude 10 units, are inclined to each other at 60°. Find the magnitude of their resultant.

3. Two perpendicular vectors have magnitudes 5 units and 12 units. Find the magnitude of the resultant and the angle it makes with the 5-unit vector.

4. Two forces of equal magnitude act at a point. If their resultant has the same magnitude as each force, find the angle between the two forces.

5. Two vectors of magnitudes 8 N and 6 N are inclined at 60°. Find the magnitude of their resultant.

6. Two vectors of magnitudes 7 units and 24 units are inclined at 120°. Find the magnitude of their resultant.

7. For two vectors **A** and **B**, the magnitude of the sum **A** + **B** equals the magnitude of the difference **A** − **B**. Find the angle between **A** and **B**.

8. Two equal vectors, each of magnitude 5 units, are inclined at 90°. Find the magnitude of their resultant and the angle it makes with either vector.

9. A person walks 4 km due north and then 3 km due east. Find the magnitude of the net displacement and its direction (as an angle east of north).

10. Two forces of 5 N and 3 N act at a point and have a resultant of magnitude 7 N. Find the angle between the two forces.

11. Two vectors of magnitudes 2 units and 3 units are inclined at 30°. Find the magnitude of their resultant.

12. Two vectors **A** and **B** have magnitudes 6 units and 10 units. Their resultant is found to be perpendicular to **A**. Find the angle between **A** and **B**, and the magnitude of the resultant.

13. Three forces acting at a point keep it in equilibrium. Two of them are 8 N and 6 N, perpendicular to each other. Find the magnitude and direction of the third force.

14. Two equal forces, each of magnitude P, are inclined at an angle θ. If their resultant has magnitude $P\sqrt{3}$, find θ.

15. Two vectors of magnitudes 4 N and 4 N are inclined at 120°. Find the magnitude of their resultant.

16. Two forces of 12 N and 5 N act at a point and give a resultant of 13 N. Are the two forces perpendicular? Justify your answer. Also find their resultant if the same two forces act in exactly opposite directions.

17. Two equal vectors, each of magnitude 3 units, are inclined at 60°. Find the magnitude of the resultant and the angle it makes with either vector.

18. Three vectors of equal magnitude are arranged so that each makes an angle of 120° with the next (they are placed symmetrically around a point). Find the magnitude of their resultant.

19. Two perpendicular forces have magnitudes 10 N and $10\sqrt{3}$ N. Find the magnitude of the resultant and the angle it makes with the 10 N force.

20. Two forces act at a point with 60° between them. One force is 6 N and their resultant is 14 N. Find the magnitude of the other force.

21. A particle undergoes two successive displacements of 6 m and 10 m, with 60° between the two displacement vectors. Find the magnitude of the net displacement.

22. Two vectors of magnitudes 3 units and 5 units give a resultant of 7 units. Find the angle between them. If this angle is then doubled, find the new magnitude of the resultant.

23. Two forces of 5 N and 8 N are inclined at 60°. Find the magnitude of the resultant and the angle it makes with the 5 N force.

24. Two forces, each of magnitude F, act at a point with angle θ between them. Find θ for which the resultant has magnitude (a) F, (b) $F\sqrt{2}$, (c) $F\sqrt{3}$, (d) 2F.

25. Two perpendicular vectors have magnitudes 9 units and 12 units. Find the magnitude of the resultant and the angle it makes with the 12-unit vector.

26. Two equal forces, each of magnitude F, act at a point at 90° to each other. A third force, also of magnitude F, acts along the bisector of the angle between them (i.e. along their resultant). Find the magnitude of the resultant of all three forces.

Detailed Solutions

Key relations used throughout

- Magnitude of resultant of **A**, **B** with angle θ between them: $R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
- Direction (angle α of the resultant from **A**): $\tan\alpha = (B\sin\theta) / (A + B\cos\theta)$
- Equal magnitudes (A = B): $R = 2A\cos(\theta/2)$, and the resultant bisects the angle
- Special cases: $\theta = 0° \rightarrow R = A + B$ (maximum) ; $\theta = 180° \rightarrow R = |A - B|$ (minimum) ; $\theta = 90° \rightarrow R = \sqrt{A^2 + B^2}$

1. $R = \sqrt{3^2 + 4^2 + 2\cdot3\cdot4\cdot\cos 90°} = \sqrt{9 + 16 + 0} = \sqrt{25}$ = **5 N**.

2. $R = \sqrt{10^2 + 10^2 + 2\cdot10\cdot10\cdot\cos 60°} = \sqrt{200 + 100} = \sqrt{300} = 10\sqrt{3} \approx$ **17.32 units**. (Or $R = 2A\cos(\theta/2) = 2\cdot10\cdot\cos 30° = 10\sqrt{3}$.)

3. $R = \sqrt{5^2 + 12^2} = \sqrt{169}$ = **13 units**. Direction from the 5-unit vector: $\tan\alpha = (12\sin 90°)/(5 + 12\cos 90°) = 12/5 = 2.4$, so $\alpha = \tan^{-1}(2.4) \approx$ **67.4°**.

4. Equal magnitudes ⇒ $R = 2F\cos(\theta/2)$. Set $2F\cos(\theta/2) = F \Rightarrow \cos(\theta/2) = ½ \Rightarrow \theta/2 = 60° \Rightarrow \theta =$ **120°**.

> ***Watch out:*** *The resultant of two equal vectors equals either one only when they are 120° apart — worth memorising.*

5. $R = \sqrt{8^2 + 6^2 + 2\cdot8\cdot6\cdot\cos 60°} = \sqrt{64 + 36 + 48} = \sqrt{148} = 2\sqrt{37} \approx$ **12.17 N**.

6. $\cos 120° = -½$. $R = \sqrt{7^2 + 24^2 + 2\cdot7\cdot24\cdot(-½)} = \sqrt{49 + 576 - 168} = \sqrt{457} \approx$ **21.38 units**.

7. $|\mathbf{A}+\mathbf{B}| = |\mathbf{A}-\mathbf{B}| \Rightarrow A^2+B^2+2AB\cos\theta = A^2+B^2-2AB\cos\theta \Rightarrow 4AB\cos\theta = 0 \Rightarrow \cos\theta = 0 \Rightarrow \theta =$ **90°**.

> ***Watch out:*** *Sum and difference are equal in length exactly when the vectors are perpendicular (the parallelogram becomes a rectangle, whose diagonals are equal).*

8. $R = \sqrt{5^2 + 5^2} = \sqrt{50} = 5\sqrt{2} \approx$ **7.07 units**. The vectors are equal, so the resultant bisects the angle ⇒ **45°** from each.

9. The two legs (north, then east) are perpendicular. Displacement = $\sqrt{4^2 + 3^2} = \sqrt{25}$ = **5 km**. Direction: $\tan\beta = 3/4 \Rightarrow \beta = \tan^{-1}(0.75) \approx$ **36.9° east of north**.

10. $R^2 = A^2 + B^2 + 2AB\cos\theta \Rightarrow 49 = 25 + 9 + 2\cdot15\cdot\cos\theta \Rightarrow 49 = 34 + 30\cos\theta \Rightarrow \cos\theta = 15/30 = ½ \Rightarrow \theta =$ **60°**.

11. $R = \sqrt{2^2 + 3^2 + 2\cdot2\cdot3\cdot\cos 30°} = \sqrt{13 + 12\cdot(\sqrt{3}/2)} = \sqrt{13 + 6\sqrt{3}} \approx \sqrt{23.39} \approx$ **4.84 units**.

12. "Resultant ⊥ **A**" means the component of the resultant along **A** is zero: $A + B\cos\theta = 0 \Rightarrow \cos\theta = -A/B = -6/10 = -0.6 \Rightarrow \theta =$ **126.9°**. The resultant then equals the part perpendicular to **A**: $R = B\sin\theta = 10\cdot0.8$ = **8 units**. (Check: $\sqrt{36 + 100 - 72} = \sqrt{64} = 8$.)

> ***Watch out:*** *The correct condition is $A + B\cos\theta = 0$, not $B + A\cos\theta = 0$ — students often resolve along the wrong vector.*

13. Resultant of the two perpendicular forces = $\sqrt{8^2 + 6^2}$ = 10 N, at $\tan^{-1}(6/8) \approx 36.9°$ from the 8 N force. For equilibrium the third force must be equal and opposite: **10 N**, directed opposite to this resultant (at 180° to it).

14. Equal forces ⇒ $R = 2P\cos(\theta/2)$. Set $2P\cos(\theta/2) = P\sqrt{3} \Rightarrow \cos(\theta/2) = \sqrt{3}/2 \Rightarrow \theta/2 = 30° \Rightarrow \theta =$ **60°**.

15. Equal vectors at 120°: $R = 2\cdot4\cdot\cos 60° = 8\cdot½$ = **4 N**. (The resultant equals each force — exactly the 120° result again.)

16. $13^2 = 12^2 + 5^2 + 2 \cdot 12 \cdot 5 \cdot \cos\theta \Rightarrow 169 = 169 + 120\cos\theta \Rightarrow \cos\theta = 0 \Rightarrow \theta = 90°$, so **yes, perpendicular**. At $\theta = 180°$: $R = |12 - 5| =$ **7 N**.

> ***Watch out:*** *12, 5, 13 is a right-triangle triple — that is the clue that the angle is 90°.*

17. $R = 2 \cdot 3 \cdot \cos 30° = 6 \cdot (\sqrt{3}/2) = 3\sqrt{3} \approx$ **5.20 units**, bisecting the angle ⇒ **30°** from each.

18. By symmetry the resultant is **0**. Reason: adding the first two (120° apart) gives a vector of magnitude $2a\cos 60° = a$, pointing exactly opposite to the third vector, so it cancels it.

19. $R = \sqrt{(10^2 + (10\sqrt{3})^2)} = \sqrt{(100 + 300)} = \sqrt{400} =$ **20 N**. Angle with the 10 N force: $\tan\alpha = (10\sqrt{3})/10 = \sqrt{3} \Rightarrow \alpha =$ **60°**.

20. Let the other force be B. $14^2 = 6^2 + B^2 + 2 \cdot 6 \cdot B \cdot \cos 60° \Rightarrow 196 = 36 + B^2 + 6B \Rightarrow B^2 + 6B - 160 = 0 \Rightarrow B = (-6 + \sqrt{676})/2 = (-6 + 26)/2 =$ **10 N**. (Reject the negative root.)

21. $R = \sqrt{(6^2 + 10^2 + 2 \cdot 6 \cdot 10 \cdot \cos 60°)} = \sqrt{(136 + 60)} = \sqrt{196} =$ **14 m**.

22. $49 = 3^2 + 5^2 + 30\cos\theta \Rightarrow \cos\theta = ½ \Rightarrow \theta =$ **60°**. Doubled: $\theta' = 120°$, $\cos 120° = -½$. $R' = \sqrt{(9 + 25 + 30 \cdot (-½))} = \sqrt{(34 - 15)} = \sqrt{19} \approx$ **4.36 units**.

> ***Watch out:*** *Doubling the angle does NOT double or halve the resultant — it must be recomputed from the formula.*

23. $R = \sqrt{(5^2 + 8^2 + 2 \cdot 5 \cdot 8 \cdot \cos 60°)} = \sqrt{(89 + 40)} = \sqrt{129} \approx$ **11.36 N**. Direction from the 5 N force: $\tan\alpha = (8\sin 60°)/(5 + 8\cos 60°) = 6.93/9 = 0.770 \Rightarrow \alpha \approx$ **37.6°**.

24. Equal forces ⇒ $R = 2F\cos(\theta/2)$. (a) $F \Rightarrow \cos(\theta/2)=½ \Rightarrow \theta =$ **120°**. (b) $F\sqrt{2} \Rightarrow \cos(\theta/2)=1/\sqrt{2} \Rightarrow \theta =$ **90°**. (c) $F\sqrt{3} \Rightarrow \cos(\theta/2)=\sqrt{3}/2 \Rightarrow \theta =$ **60°**. (d) $2F \Rightarrow \cos(\theta/2)=1 \Rightarrow \theta =$ **0°** (vectors parallel).

25. $R = \sqrt{(9^2 + 12^2)} = \sqrt{225} =$ **15 units**. Angle with the 12-unit vector: $\tan\alpha = 9/12 = 0.75 \Rightarrow \alpha =$ **36.9°**.

26. The two equal forces at 90° give a resultant of $F\sqrt{2}$ along their bisector (45° from each). The third force F lies along this same bisector, so it simply adds on: total $= F\sqrt{2} + F = F(\sqrt{2} + 1) \approx$ **2.41 F**, along the bisector.

> ***Watch out:*** *Because the third force is collinear with the resultant of the first two, the magnitudes add directly — no cosine rule needed for the last step.*